006-MoPr02 · TeleField for Zoho CRM · Portfolio rank 2 (WIN order)

TeleField — Product Architecture & Build Status

What the product is, what is finished, what is not, and precisely why not

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Companion	claude_V001_MoPr02-telefield_zoho_versiondetail_manualtest_20260920_1232.xlsx
Classification	Internal — product status, not customer-facing copy

1. Document control

Version	Date / time	Author	Change description	Status
V001	2026-09-20 12:32	Claude for Senthil/KP	First issue. Pictorial explanation of TeleField built against code release v033: four figures (request path and its unconnected edges; the compliant call lifecycle; the release history; the twelve acceptance gates), the completed inventory with the release that delivered each item, the not-completed inventory with the blocking reason and its owner, and the decisions awaiting ratification. Supersedes no earlier document.	Released
Next issue V002 is due when any of PX07–PX09 lands, because each of those changes a figure: PX07 verifies the two platform assumptions, PX08 connects the provider edge, PX09 connects the notification edge.				

Approval

Role	Name	Action	Signature / date
Product owner / founder	Senthil (KP)	Approve & ratify P-D1…P-D9	
Author	Claude	Prepared	2026-09-20
Second reviewer (DR-12)	*to be nominated*	Walk the pack, sign	

2. Context

What TeleField is. A Zoho CRM Marketplace extension for Indian SME sales floors. A rep opens a widget inside Zoho CRM; TeleField hands them the next lead to call, records the outcome and the disposition, schedules callbacks, plans the day's field visits (beats) with geo-verified check-ins, scores the team, and bills the tenant. It is built in TypeScript/Node with an Express HTTP surface, server-rendered screens and Postgres persistence.

Where the product sits. Three independent reviews have now been run against it. The first (internal, 17 cold passes, v013–v031) found 35 defects in code that already carried 100% test coverage. The second (ChatGPT, 14 Sep, against v031) returned *BUILD, NOT LAUNCH* with 10 findings and a 13-sprint order E01–E13. The third (Perplexity, 16 Sep, against v032) returned 8 further findings, downgraded two acceptance gates, added two more, and restructured the work into 14 sprints PX01–PX14. Release v033 executes every one of those sprints that can be finished without something from outside this environment.

The one fact that governs everything below. Every remaining piece of work needs an account, an org, a vendor or a customer. None of it is blocked on engineering effort. That is a scheduling problem in Senthil's calendar, not a backlog problem in the code — and it is why a product with 1,769 passing tests still cannot place a telephone call.

3. How to read this document

- **§4 Figure 1** is the one to read if you read only one thing: it draws a single dial from the rep's click to the handset, shows every check it passes through, and shows the four edges that are drawn dashed because nothing is connected to them.
- **§5 Figure 2** is for anyone asking "what happens to a call" — the state machine that PX04 made explicit.
- **§6 Figure 3** is the history: 33 releases in four phases, and the sprints that remain.
- **§7 Figure 4** is the launch scoreboard: twelve acceptance gates and their status.
- **§8** lists what is complete and which release delivered it. **§9** lists what is not, with the reason and the owner. **§10** is what needs a decision from Senthil.
- **Colour means the same thing in every figure:** see the key immediately below. Every figure is also available as a standalone `.svg` beside this document, so it can be dropped into a deck.

■ BUILT & PROVEN in this environment, pinned by tests ■ BUILT, UNPROVEN LIVE correct in code, never run against the real platform ▨ NOT BUILT needs an account, org, vendor or customer ● LAUNCH BLOCKER

4. The product in one picture: a single dial, end to end

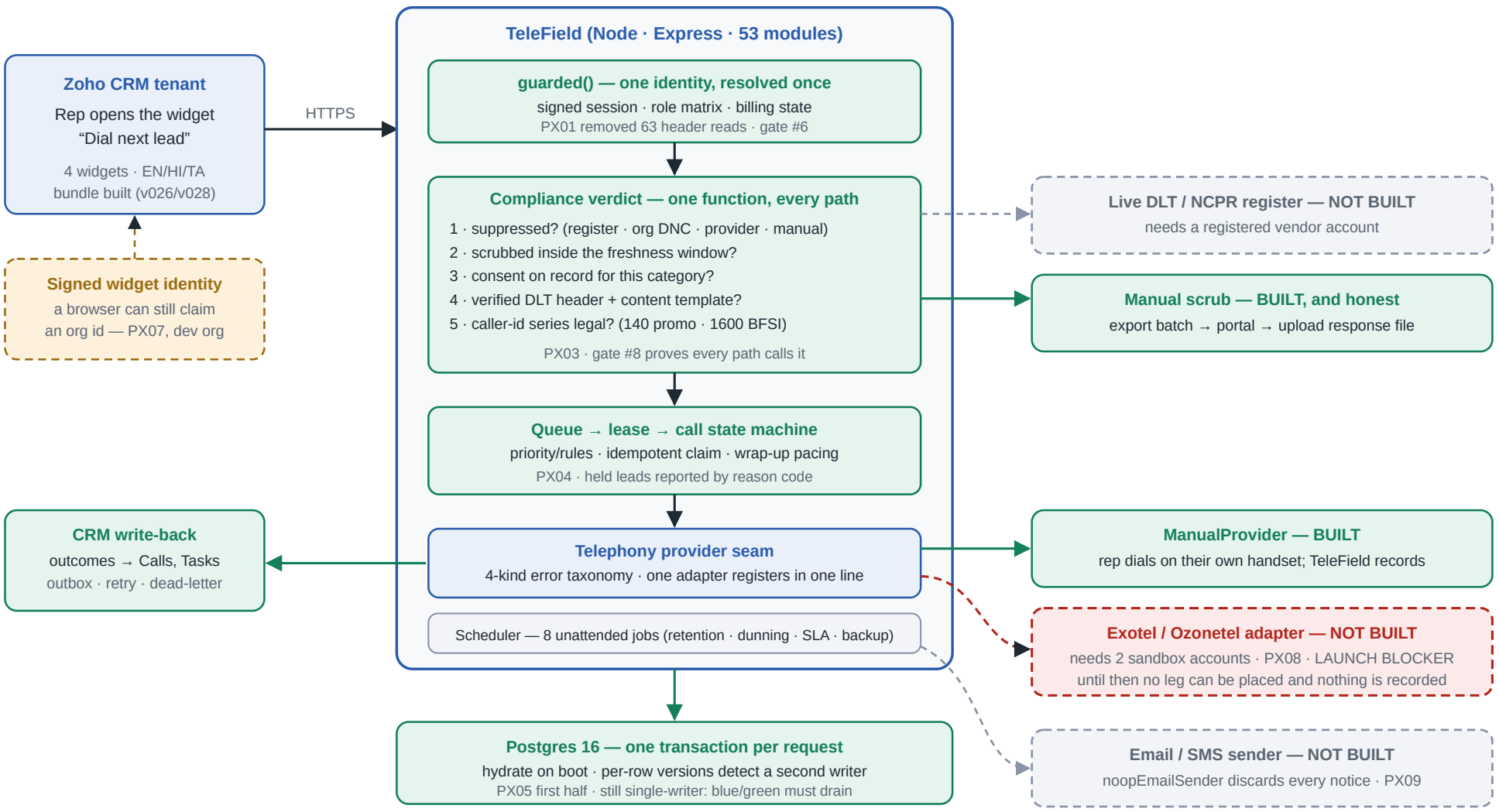


Figure 1 — One dial, end to end. Everything inside the blue box is built and pinned by tests. The rep's dial passes identity, then five compliance checks, then the queue, and arrives at a seam where exactly one provider is connected: the rep's own handset. The four dashed edges are the whole of what is missing — a telephony vendor, a register vendor, a notification vendor, and one Zoho developer org — and not one of them is an engineering task.

Read the picture this way. TeleField today is a *complete, compliant, audited workflow around a call that the rep places by hand.* That is a real product for a floor that already dials manually, and it is not the product the listing describes. The gap between the two is one dashed box.

5. What happens to a call

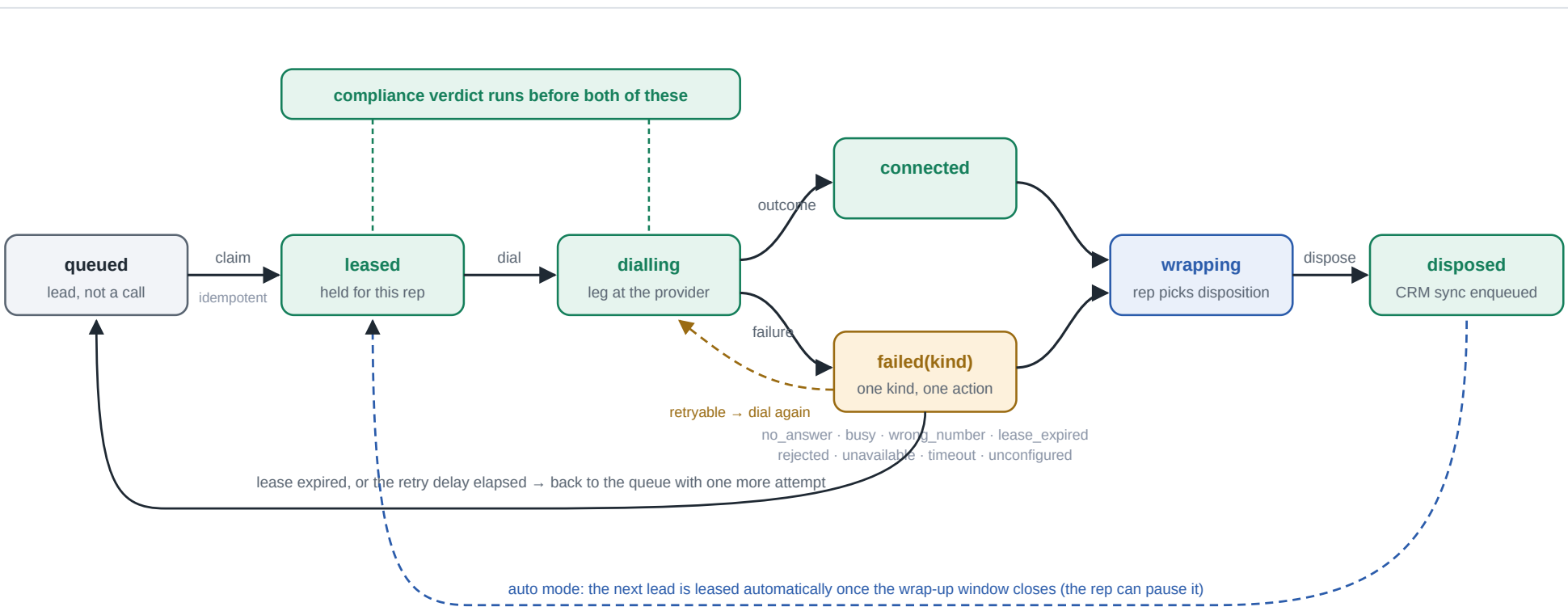


Figure 2 — The call lifecycle (PX04). Before v033 a call's state was inferred from two nullable fields in four different places. It is now explicit: every transition is idempotent, every invalid one is a named error, and every provider failure maps to exactly one action the rep can read. A crash in any of these seven states is tested — the lead is never lost, never leased twice, never silently stuck.

6. How it was built, and what remains

v001 – v012 · BUILD
10–11 Sep · 12 sprints
98 of 100 stories in code
1,261 tests at 100%
labelled “FINAL” at v012

v013 – v031 · COLD REVIEW
13 Sep · 17 adversarial passes
35 defects · 16 artefact gaps
Postgres · screens · extension
all inside code that already had 100% coverage

v032 · CHATGPT REVIEW
15 Sep · “BUILD, NOT LAUNCH”
10 findings · E01–E13
widget became real (E01)
2 blockers closed

v033 · PERPLEXITY REVIEW
20 Sep · 8 findings · PX01–PX14
PX01–PX06 executed
1,769 tests · 9 gates
identity · compliance · money · state

Each review found defects the previous one and 100% coverage had both missed.
Three of the four Perplexity findings were **recurrences of classes already closed once** — which is why every sprint now ships the gate, not just the fix.

What remains — 8 sprints, and what each one is waiting for

PX08 · place a real call
2 provider sandbox accounts
owner: Senthil · blocks launch

PX07 · live install
Zoho developer account + dev org
owner: Senthil · blocks launch

PX09 · notifications
email / SMS provider account
owner: Senthil · blocks charging

PX13 · willingness to pay
3–5 design partners
owner: Senthil · do this first

PX12 · procurement
external pen-test vendor
owner: Senthil

PX14 · AI follow-up
needs recordings + pilot signal
correctly last

PX10 · mobile + control tower
nothing blocking — Claude work
next on “go ahead”

PX11 · vernacular + vertical
nothing blocking — Claude work
next on “go ahead”

Figure 3 — Thirty-three releases, and the eight sprints left. Six of the eight are waiting on something only Senthil can obtain; two are ordinary engineering work that simply was not in this release's scope. The review's own recommendation is to run PX13 (five partner conversations) *before* PX08, because PX08 is the most expensive and least reversible sprint in the plan.

7. The launch scoreboard

Acceptance gate		Status after v033	Unblocked by
Core call workflow	FAIL — cannot place a leg	state machine + gate built; no provider connected	PX08
Tenant identity	PARTIAL — code fixed	PX01 closed the cross-user defect; live proof outstanding	PX07
Zoho live compatibility	UNPROVEN	two named platform assumptions never verified	PX07
Durability	PARTIAL — transactional	one transaction per request; still single-writer	PX05 (2nd half)
Privacy / DNC	PARTIAL — suppression live	per-number suppression, consent ledger, erasure parity	register vendor
Observability	PARTIAL — measured, not sent	every alert is computed and then discarded	PX09
Security	PARTIAL — no outside attack	strong static controls; nobody external has attacked it	PX12
Marketplace listing	OPEN	nothing submitted; needs the dev org and the assets	PX07
Customer validation	OPEN — zero users	highest-value unblocked item in the whole plan	PX13
Differentiation	AT RISK	Zoho now ships telephony, a power dialler and RouteIQ	PX13
Supply chain / clean build	PASS in code	a clean clone with no cache builds, tests and renders — proven from the zip	done (PX02)
Regulatory fitness (India)	PARTIAL — every local check built	the live register is the only missing piece	register vendor

Ten of twelve gates need evidence that cannot be produced inside this environment. That is the finding, not a complaint about the code.

Figure 4 — Twelve acceptance gates. One passes in code. Six are partial. Five are open or failing. No gate reaches PASS on static grounds, because a gate marked PASS without live evidence is exactly the failure the reviews charged the product with.

8. What is complete

Every row below is built, wired into a production call path, and pinned by tests that fail if it is removed. 44 suites / 1,769 tests at 100% statements, branches, functions and lines, zero coverage pragmas, verified from a clean clone of the shipped zip.

Capability	Release	What it actually does	Proof
Install & lifecycle	v001, v015–v017, v025	OAuth install with CSRF state, uninstall wipes tokens and schedules deletion, reinstall restores inside retention, rotation-safe refresh, webhook HMAC + replay guard, per-org data centre.	S1 + review passes 3, 5, 11
Call queue & dispositions	v002–v004	One-screen setup on defaults, priority/FIFO ordering with boost/deprioritise/skip rules, retries, callbacks that jump the queue, custom disposition taxonomy, CRM sync outbox with named errors.	S2–S4 suites
Beat planner & geo check-ins	v005–v006	Recurrence-aware daily routes, cooldowns, nearest-neighbour ordering with per-leg distance, geofenced check-in/check-out with reject/flag/allow policy and sequence enforcement.	S5–S6 suites

Billing, invoices, GST	v006–v007, **v033**	14-day trial without a card, exact mid-cycle proration, 7-day grace with dunning at day 0/3/6, read-only mode without data loss, plan limits, cancellation with save-offer. v033 added integer minor units with a currency on every amount, INR by default, and a GST invoice that splits CGST/SGST or IGST exactly.	S6–S7 + PX06 suite
Manager scorecards & analytics	v008	Six KPIs from data already held, per-rep cards with grades, ranked team view, trends, activation funnel, health score, CSV export.	S8 suite
Privacy & retention	v009, v013, v016, v023	Self-serve export and delete-my-data with per-dataset erasure evidence, 30-day retention sweep that actually runs on a schedule, daily backups written to disk and integrity-checked before being reported, audit rows anonymised rather than deleted.	F-T01/T02/T17/T25 regressions
Nine screens, WCAG AA	v019	Onboarding, call workspace, rule builder, planner, mobile route, settings, billing, analytics, help — each rendered from the same view-model its JSON route serves, audited on every render.	GATE-4 + a11y engine
Zoho extension bundle	v026, v028	plugin-manifest.json, four widget bootstraps, EN/HI/TA translation catalogues, packaged by `npm run extension` — the thing a customer actually installs, which did not exist until pass 12.	GAP-T14/T16
Postgres persistence	v018, **v033**	Full adapter over the Store interface, hydrate on boot, flush before responding. v033 made each flush one transaction and added per-row versions that detect a second writer.	58 DB-backed tests, 100% on the adapter
Resilience	v023, v029–v031	Scheduler with 8 unattended jobs, database-outage degradation (holds writes, refuses to promise durability, reconnects and replays in order), SIGTERM drain, operational-table pruning so a restart stays bounded.	passes 15–17
A widget that works	v032	Read-only peek, deliberate idempotent claim, lease expiry the client is told about, bound handlers, one shared UI/API schema. Before this the controls were a picture.	CG-TF-001/002/003
Identity, once	**v033**	The signed session is resolved in one place and handed to every handler; the development header is read in exactly one file and never honoured in production.	gate #6 + production-mode suite
India telecom compliance	**v033**	Consent ledger, DLT header/template/entity registry, suppression keyed on the number, scrub-before-release with a freshness window, 140/1600 caller-id policy by campaign category, per-attempt traceability, a regulator report, and a 15-day suspension stub. One verdict function, called from every path that can put a number in front of a rep.	gate #8 + 51 tests
Real auto-dial	**v033**	Explicit call state machine, progressive dialling with a server-enforced wrap-up window and a per-rep pause, durable outbox whose attempt counter survives a re-enqueue, a visible and replayable dead-letter queue.	chaos suite, 7 states
Nine standing gates	v022, **v033**	Reachability, published claims, public surface, screens, evidence, identity, dependencies, compliance, money — plain Node, portable by config, and each one mutation-tested by breaking the thing it watches.	`npm run gates`

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9. What is not complete, and why

Ordered by what it blocks. "Why not" is the real reason, not a euphemism for "not got to it yet".

Missing	Sprint	Why it is not built	What unblocks it	Consequence today
Place a real phone call LAUNCH BLOCKER	PX08	An adapter needs a provider sandbox to be written against. Writing one from published documentation alone produces code that is self-consistent and wrong about the platform — which is precisely what VER-T1, VER-T4 and VER-T5 cost this product three separate times. The seam, the error taxonomy and the compliance gate in front of it are all built; the adapter is deliberately not guessed.	Two India provider sandbox accounts (Exotel / Ozonetel class) — Senthil, decision D18	The product records calls the rep places on their own handset. The listing's central promise is unmet.
Live install in a Zoho org LAUNCH BLOCKER	PX07	Two platform assumptions (the CRM service name and the SDK URL) are isolated in one file and flagged as unverified. Install, uninstall, disable, enable and update webhook shapes, scope approval, Multi-DC behaviour and the marketplace validator all need a real org to exercise.	Zoho developer account + Sigma project + dev org — Senthil, LC-1	Nobody has ever installed TeleField. A browser can still assert an org id; the server only checks that the org exists.
Live DLT / NCPR register	PX03 remainder	Connecting to the register requires a registered vendor relationship. Everything local was built instead: the consent ledger, the registry, the suppression table, the freshness gate and the report. The manual path — export the batch, scrub on the portal, upload the response — is functional and honest, exactly as ManualProvider is for telephony.	A register/DLT vendor account — Senthil	Scrubbing is an operator task on a schedule rather than an automatic call. The gate still refuses to dial an unscrubbed number.
Notifications that reach a human	PX09	No email or SMS provider is connected, so `noopEmailSender` is wired into the production scheduler. Every dunning notice, SLA breach, error spike and scheduler failure is computed correctly and then discarded.	An email/SMS provider account — Senthil	**Do not take money before this ships.** A paying tenant whose card fails is never told, and neither is the operator.

Call recording	PX08	Recording is produced by the telephony provider; there is nothing to record without an adapter. The pieces it will need — consent basis with an evidence reference, per-attempt traceability, erasure parity for new datasets — were built in PX03 so it is not a retrofit.	Rides PX08	The AI roadmap (PX14) has no input, and BFSI buyers cannot be answered on compliance recording.
Async DB-first repository	PX05 2nd half	The `Store` interface is synchronous and all 53 modules are built on it; making it async is a rewrite of every call site, not a sprint item. v033 took the half that could be taken safely — transactional batches and conflict detection — and stopped there rather than starting a rewrite it could not finish.	Nothing external — scheduled Claude work	Still single-writer: blue/green deploys must drain the old process first. A crash between a mutation and its flush loses that one in-flight request.
Mobile PWA + manager control tower	PX10	Not blocked at all — simply outside this release's scope, which the review itself set at PX01–PX06. The server-side halves (offline sync API with dedupe, scorecards, metrics, funnels, SLOs) already exist.	Nothing — say "go ahead"	Field reps have no offline shell; managers have the numbers but not the two-minute drill-down.
Vernacular screens + vertical pack	PX11	Also unblocked and out of scope. The manifest declares en/hi/ta and the widget chrome is translated; the nine screens are English, which the code states honestly rather than pretending otherwise.	Nothing — say "go ahead"	"Vernacular" is a weak claim as sold. Three strings in Tamil is not a vernacular product.
SSO/SCIM, pen-test, DR proof	PX12	An external penetration test needs a vendor engagement. The static controls are strong and scanned in CI, but nobody outside this build has ever attacked the product.	Pen-test vendor — Senthil	Enterprise procurement cannot be answered. Scope PX01's finding into the brief when it is written.
Evidence anyone wants it HIGHEST VALUE	PX13	No design partner has ever used the product. This needs five conversations, not code — and the review is explicit that it should happen *before* PX08, because PX08 is the most expensive and least reversible sprint in the plan.	3–5 design partners, two in a DLT-heavy vertical — Senthil	Zero users. The repositioning (P-D1) is a hypothesis with code behind it and no market evidence.
AI follow-up intelligence	PX14	Correctly last: it depends on recordings (PX08) and on the pilot saying the workflow is wanted (PX13). Building it now would be building on two unvalidated assumptions.	PX08 + PX13	None — this is the right thing to not have built.

> **The pattern worth naming.** Nine of the eleven rows above are blocked by an account, an org, a vendor or a customer. The build loop has been optimising for artefacts that can be produced without external dependencies, and it has produced them to an unusually high standard. Every remaining gate needs something from outside. That is a sequencing problem in the calendar, not a backlog problem in the code.

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10. Waiting on a decision

ID	Decision	Recommendation and why
P-D1	Reposition to "TRAI-compliant outbound + field execution on Zoho CRM"?	**Yes.** Zoho now ships built-in telephony, a power dialler and RouteIQ first-party; the original wedge is partly their product now. The compliance-native wedge is already built in code.
P-D2	Accept compliance as a P0 sprint ahead of the adapters?	**Yes** — executed in v033. 140/1600 enforcement has been mandatory since 1 January 2026.
P-D3	Adopt the PX ordering, keeping the E-mapping?	**Yes.** The tracker reconciles rather than resets.
P-D4	Move the design-partner sprint before the telephony adapters?	**Yes.** Validate the repositioned pitch before spending the most expensive sprint.
P-D5	Verify the real Zoho marketplace revenue share and settlement currency?	**Yes, before pricing.** Zoho's own FAQ states no rate; the 15% assumption is not confirmed anywhere.
P-D6	Remove the inert "auto" setting, or build it?	**Resolved** — built in v033, so nothing shipped visible-but-inert.
P-D7	Freeze feature work until identity and the clean build land?	**Yes** — honoured: PX01 and PX02 went first.
P-D8	Default compliance category for an org that has declared none?	Built as **transactional** (the narrowest: no scrub, consent or 140/1600 obligations, and it cannot run a promotional campaign until it declares one). The stricter alternative — refuse every dial until a category is declared — breaks the one-screen setup promise. **Your call.**
P-D9	Ratify INR prices: Starter ₹1,499/mo (3 users), Pro ₹4,999/mo (25 users)?	Proposed against the live ₹99–₹1,349 per-user band; these sit inside it at ₹500 and ₹200 per user. One line to change. Decide after P-D5.

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11. Glossary

Term	Meaning in this document
DLT	Distributed Ledger Technology platform operated under TRAI rules, on which every commercial caller registers its entity, sender headers and content templates before running campaigns.
NCPR	National Customer Preference Register — the do-not-disturb register outbound lists must be scrubbed against before each campaign run.
140 / 1600 series	Number ranges mandated since 1 January 2026: 140 for promotional calls, 1600 for BFSI service calls.
Suppression	A block held against a phone *number* (not a lead record), so a re-imported contact stays blocked.

Lease	The period a claimed lead is held for one rep before it returns to the queue.
Gate (review gate)	A build-breaking check that catches a *class* of defect rather than one instance. Nine of them run as `npm run gates`.
Gate (acceptance gate)	One of the twelve launch criteria in Figure 4. Unrelated to the above, and unfortunately both are called gates.
PX01–PX14	The sprint numbering from the Perplexity review, mapped onto the earlier E01–E13 numbering from the ChatGPT review.
Done-in-code	Built and tested here, but with no live evidence. It is deliberately not called "done".

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12. Source documents

- `perplexity_review_V001_MoPr02-telefield_zoho_20260916_1813.md` — the third review, and the source of PX01–PX14, P-01–P-08 and P-D1–P-D7.
- `claude_v033_telefield_sprint_tracker_traceability_20260920_1014.xlsx` — the live tracker: 140 execution events, 114 findings, 100-story traceability matrix, twelve acceptance gates, RAID.
- `claude_v033_telefield_px01_px06_perplexity_review_code_20260920_1014.zip` — the code this document describes; `docs/REVIEW_PACK.md` inside it is the reviewer's entry point.
- `claude_V001_MoPr02-telefield_zoho_versiondetail_manualtest_20260920_1232.xlsx` — the companion: every version with its rationale, the pending versions with their reasons, step-by-step execution detail and the manual test pack.